



Ahmed Maghri

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Education:

B.Sc. Computer Science (CS), Constructor University Bremen, Germany.

Bremen, Germany | Sep 2024 – Jun 2027 (expected)

- **Minor:** Mathematics & Modeling.
 - **GPA:** 1.49 (German scale, where 1.0 is the best).
 - **Relevant Courses:** Algorithms and Data structures: 100% | Computer Networks: 97% | Programming in C & C++: 99% | Matrix Algebra and Advanced Calculus: 98% | Probability and Random Processes: 100% | Discrete Mathematics: 100%
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Work Experience:

- **Research Intern in Semantic Communications & AI for Next-Generation Networks** | *York University*
Toronto, Canada | Jun 2026 - Present
 - Selected as one of ~60 German students nationwide for a **EUR5,000 MITACS & DAAD RISE Worldwide** fellowship.
 - Investigating state-of-the-art semantic communication systems and AI-driven approaches for next-generation networks; prototyping and benchmarking novel methods under Prof. Ping Wang (**IEEE Fellow**, York Research Chair, **30,000+ citations**).
 - **Software Engineer (Werkstudent)** | *Insytes GmbH*
Heilbronn, Germany | Feb 2026 – Jun 2026
 - Implemented causal feature permutation to identify top causal factors without running full PCMC/FCI, enabling fast preliminary results while the full analysis runs in parallel.
 - Diagnosed an AWS runtime regression in the causal pipeline by tracing a CPU/thread allocation mismatch in ECS; tuned task resources to eliminate throttling, cutting runtime from ~6 min to ~1 min.
 - Researched causal AI methods for unstructured data and presented findings to a committee of the **Samsung Research & Innovation department**, including Natasha Jaeger.
 - **Teaching Assistant in Programming in C/C++ and Math Foundations of Computer Science** | *Constructor University*
Bremen, Germany | Sep 2025 - Feb 2026
 - Led classes and office hours for 350+ students across C/C++ and Mathematical Foundations of CS; redesigned sessions around motivation-first explanations and live problem-solving, growing attendance from ~10 to 40-45 students/session.
 - Reduced course drop rate by 10% through structured catch-up sessions and targeted support for at-risk students.
 - **Research Student Intern in AI for Geometry Reasoning (Sengoku)** | *Constructor Tech*
Bremen, Germany | Jul 2025 - Sep 2025
 - Designed a geometry reasoning model mimicking AlphaProof's architecture: built a custom DSL for shape encoding and trained on high-school-level geometry problems, achieving end-to-end problem solving on a subset of benchmark cases.
 - **Backend Django Lead Intern** | *GBCS Group (fleet management firm)*
Calgary, Canada | Jun 2025 - Aug 2025
 - Flattened deeply nested REST API endpoints and reduced endpoint sprawl, cutting frontend debug time from 2–3 days to same-day resolution.
 - Led an 11-person backend team: ran SCRUM ceremonies, coordinated delivery, and introduced agile methodology one month after the API rearchitecture, combined effect doubled overall software delivery speed.
 - Optimized Scope 1–4 CO₂ calculations using NumPy and pandas vectorization, cutting average runtime from 10s to 1s.
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Projects:

- **Muslim Students Association Campus Platform**
Bremen, Germany | Jun 2025 - Present

- Co-built a 400+ user campus platform (React/Vite, Express, Firebase, Stripe) with accounts, dashboard, calendars, RSVP, and donation flows.
 - Owned the backend and Stripe donation system post-launch, growing weekly donations from €20 to €60 and average event attendance from 30 to 50.
 - **AutoAgent | 2nd Place, Constructor Knowledge Lab Hackathon**
Bremen, Germany | Nov 2025
 - Collaborated with a teammate to build AutoAgent - a tool converting research paper links into runnable demos; owned the full AI pipeline using prompt engineering, multi-agent strategies, and heuristics
 - Drove runtime from 20 minutes to 70 seconds across heuristics-only, LLM-only, and hybrid strategies, earning 2nd place at the Constructor Knowledge Lab Hackathon.
 - **Talenza | BMW x Lovable x Constructor Tech x CKL Hackathon**
Bremen, Germany | Mar 2026
 - Owned the AI pipeline end-to-end: built a 3-stage n8n agentic workflow: CV parsing and AI profile generation via Claude API, scenario definition via Q&A agent, and multi-dimensional scoring across 5 weighted factors.
 - Platform re-ranks the same candidate pool differently per business scenario (Crisis, Transformation, Growth, Succession) and compares internal vs. external hiring paths.
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Extracurricular activities:

- **President & Undergrad Research Supervisor, Mathematics Society Constructor University**
Bremen, Germany | Aug 2025 - Present
 - Collaborated with professors in **supervising 9 undergraduate research projects**, including *Runge-Kutta Methods in Neural Networks*, *Paxos Algorithm & Distributed Systems*, *Drone Irrigation Optimization*, *Benford's Law*, *Godel's Incompleteness Theorems*, and *Fundamental Theorem of Arithmetic & Algebra*.
 - Co-launched the first inter-school Olympiad at Constructor University with the Mathematics Faculty; **managed a 28-person cross-functional team** (logistics, tech, marketing, competitive math, outreach, audiovisual) for end-to-end execution.
 - Founded and led the "Math Behind ..." event series; delivered 3 talks including **"Math Behind AI/ML/Neural Networks"**, **"Math Behind Cryptography/Cybersecurity/Quantum Computing"**, and **"The Numbers Behind Trade: When Math Meets Wall Street"**.
 - Organized 20+ events with 50 attendees/event over 1.5 years, including faculty talks with PhD professors and a research contest **Datathon** in collaboration with Insytes GmbH and the Mathematics Faculty.
 - **Intercultural Trainer | International Office & Psychological Counseling, Constructor University**
Bremen, Germany | Sep 2025 – Present
 - Co-designed and facilitated 3 intercultural training sessions for groups of 30 students, covering cultural adaptation, identity, and conflict resolution.
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Technical Skills

- **Programming Languages:** C, C++, Python, Kotlin, Assembly, JavaScript, HTML, CSS.
 - **Frameworks:** Django, React, Express, Node/Vite, FastAPI.
 - **Machine Learning & AI:** Causal ML (PCMCI, FCI), NLP, Prompt Engineering, Multi-Agent Systems, n8n.
 - **Database:** SQL, MySQL, PostgreSQL, MongoDB
 - **Tools:** Git, Docker, Linux, AWS (ECS, Cognito, Fargate), Firebase, Stripe, Postman, ClickUp.
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Honors and Achievements:

- **German Winter Programming contest 2025:** Top 3%
 - **Bremen Google Ideathon 2025:** 1st place. Pitched a platform to eliminate middlemen in agriculture supply chains.
 - **Constructor University Pi Day Integration Bee (2026):** 1st place (duo team)
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Languages: Arabic (Native), Darija (Native), French (C2), English (C2), German (B1), Tamazight (Advanced).